

For the planar fluid motion deriving from the velocity potential

$$\varphi = M \cos(\Omega t) \cos(\theta)/r,$$

obtain:

1. Radial and azimuthal velocity components
2. Expansion rate $\nabla \cdot \vec{v}$
3. Vorticity
4. Acceleration of the fluid particle initially located at $r = 1$ and $\theta = \pi/2$
5. Initial value of the circulation around a circle of radius $R = 1/2$ centered at $(r = 1, \theta = \pi/2)$.
6. Streamlines.
7. Trajectories and particle paths.

$$1) \quad v_r = \frac{\partial \varphi}{\partial r} = -M \frac{\cos(\Omega t) \cos \theta}{r^2}$$

$$v_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = -M \frac{\cos(\Omega t) \sin \theta}{r^2}$$

$$2) \quad \nabla \cdot \vec{v} = \frac{1}{r} \left[\frac{\partial (v_r r)}{\partial r} + \frac{\partial v_\theta}{\partial \theta} \right] = 0$$

$$3) \quad \nabla \wedge \vec{v} = \omega_z \vec{e}_z = \frac{1}{r} \left[\frac{\partial (r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta} \right] \vec{e}_z = 0$$

$$4) \quad \frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + \nabla \left(\frac{v^2}{2} \right) - \vec{v} \wedge (\nabla \wedge \vec{v}) = \left(\frac{\partial v_r}{\partial t} + \frac{\partial}{\partial r} \frac{v^2}{2} \right) \vec{e}_r + \left(\frac{\partial v_\theta}{\partial t} + \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{v^2}{2} \right) \right) \vec{e}_\theta = \left[\frac{2M \sin(\Omega t)}{r^2} \cos \theta - \frac{2M^2 \cos^2(\Omega t)}{r^5} \right] \vec{e}_r + \left[M \Omega \sin(\Omega t) \sin \theta \right] \vec{e}_\theta$$

$$v^2 = \frac{M^2 \cos^2(\Omega t)}{r^4}$$

$$r=1, \theta=\frac{\pi}{2} \Rightarrow \boxed{\frac{D\vec{v}}{Dt} = -2M^2 \cos^2(\Omega t) \vec{e}_r + M \Omega \sin(\Omega t) \vec{e}_\theta}$$

$$5) \quad \Gamma = \oint \vec{\omega} \cdot \vec{e}_z ds = 0$$

$$6) \quad \frac{dr}{-M \frac{\cos(\Omega t) \cos \theta}{r^2}} = \frac{r d\theta}{-M \frac{\cos(\Omega t) \sin \theta}{r^2}} \Rightarrow \frac{dr}{r} = \frac{\cos \theta}{\sin \theta} d\theta$$

$$\boxed{r = \frac{r_0}{\sin \theta} \sin \theta}$$

$$7) \quad \left. \begin{array}{l} \frac{dr}{dt} = -M \frac{\cos(\Omega t) \cos \theta}{r^2} \\ r \frac{d\theta}{dt} = -M \frac{\cos(\Omega t) \sin \theta}{r^2} \end{array} \right\} \frac{dr}{r d\theta} = \frac{\cos \theta}{\sin \theta} \Rightarrow \boxed{r = \frac{r_0}{\sin \theta} \sin \theta} \rightarrow \text{STREAMLINES} \equiv \text{PATH LINES} !!$$

$$\frac{d\theta}{dt} = -M \frac{\cos(\Omega t) \sin^3 \theta}{r_0^3 \sin^2 \theta} \Rightarrow \int_{\theta_0}^{\theta} \sin^2 \theta d\theta = -M \frac{\sin^3 \theta_0}{r_0^3} \int_0^t \cos(\Omega t) dt$$

$$\text{T2} \quad \boxed{\frac{\theta - \theta_0}{2} - \frac{\sin(2\theta) - \sin(2\theta_0)}{4} = -\frac{M \sin^3 \theta_0}{r_0^3 \Omega} \sin(\Omega t)}$$